Task 1: CGPA Calculator

CodeAlpha C++ Internship | Marathi + English explanation

Program kay karto? (What it does)

• Student cha semester GPA ani overall CGPA calculate karto (4.0 scale).
• Pratyek course sathi grade ani credit hours input ghto, mag result table dakhavto.

Concept / Formula

• Grade Points = Grade Point x Credit Hours (pratyek course sathi).
• Semester GPA = Total Grade Points / Total Credits.
• CGPA = Sarva semesters che Total Points / Sarva semesters che Total Credits.

Code kasa kam karto? (How the code works)

• struct Course: course cha name, grade, gradePoint ani credits store karto.
• gradeToPoint(): letter grade (A, B+, C...) la number madhe convert karto. Chukicha grade asel tar -1 return karto.
• readInt() / readCredits(): input validation - chukicha input aala tar parat vicharto (cin.clear + cin.ignore).
• main(): semesters cha outer loop ani courses cha inner loop. Points ani credits add hot jatat.
• Shevti setw() ani setprecision(2) vaparun sundar table print hoto.

Video madhe kay bolaych (Video tips)

• Pahile problem sanga, mag formula samjava.
• Live demo dya: 2-3 courses enter kara ani result dakhava.
• Invalid grade (jase 'Z') takun validation dakhava.

Kasa run karaycha (How to Run)

```
g++ -std=c++17 CodeAlpha_CGPA_Calculator.cpp -o cgpa
./cgpa            (Windows: cgpa.exe)
```

Full C++ Code

```
// CodeAlpha C++ Internship - Task 1: CGPA Calculator
#include <iostream>
#include <iomanip>
#include <vector>
#include <string>
#include <limits>
#include <cctype>
using namespace std;

struct Course {
    string name;
    string grade;      // e.g. A, B+, C
    double gradePoint; // 4.0 scale
    double credits;
};

// Convert letter grade to grade point (4.0 scale). Returns -1 if invalid.
double gradeToPoint(string g) {
    for (char &c : g) c = toupper(c);
    if (g == "A+" || g == "A") return 4.0;
    if (g == "A-") return 3.7;
    if (g == "B+") return 3.3;
    if (g == "B")  return 3.0;
    if (g == "B-") return 2.7;
    if (g == "C+") return 2.3;
    if (g == "C")  return 2.0;
    if (g == "C-") return 1.7;
    if (g == "D")  return 1.0;
    if (g == "F")  return 0.0;
    return -1;
}

int readInt(const string &msg, int minV) {
    int x;
    while (true) {
        cout << msg;
        if (cin >> x && x >= minV) return x;
        cout << "  Invalid input. Enter a number >= " << minV << ".\n";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
    }
}

double readCredits(const string &msg) {
    double x;
    while (true) {
        cout << msg;
        if (cin >> x && x > 0) return x;
        cout << "  Invalid input. Credit hours must be > 0.\n";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
    }
}

int main() {
    cout << "====================================\n";
    cout << "         CGPA CALCULATOR\n";
    cout << "====================================\n";
    cout << "Grades: A, A-, B+, B, B-, C+, C, C-, D, F (4.0 scale)\n\n";

    int semesters = readInt("Enter number of semesters: ", 1);

    double overallPoints = 0, overallCredits = 0;
    vector<double> semGPA;

    for (int s = 1; s <= semesters; s++) {
        cout << "\n----- Semester " << s << " -----\n";
        int n = readInt("Number of courses: ", 1);
        vector<Course> courses;
        double semPoints = 0, semCredits = 0;

        for (int i = 1; i <= n; i++) {
            Course c;
            cout << "\nCourse " << i << " name: ";
            cin >> ws;
            getline(cin, c.name);
            while (true) {
```

```
            cout << "Grade: ";
            cin >> c.grade;
            c.gradePoint = gradeToPoint(c.grade);
            if (c.gradePoint >= 0) break;
            cout << "  Invalid grade! Try again.\n";
        }
        c.credits = readCredits("Credit hours: ");

        semPoints  += c.gradePoint * c.credits;   // grade x credit hours
        semCredits += c.credits;
        courses.push_back(c);
    }

    double gpa = semPoints / semCredits;
    semGPA.push_back(gpa);
    overallPoints  += semPoints;
    overallCredits += semCredits;

    // Display individual course grades
    cout << "\nSemester " << s << " Report\n";
    cout << left << setw(22) << "Course" << setw(8) << "Grade"
         << setw(10) << "Credits" << "Points\n";
    cout << string(48, '-') << "\n";
    for (const Course &c : courses) {
        cout << left << setw(22) << c.name << setw(8) << c.grade
             << setw(10) << fixed << setprecision(2) << c.credits
             << c.gradePoint * c.credits << "\n";
    }
    cout << string(48, '-') << "\n";
    cout << "Total Credits : " << semCredits << "\n";
    cout << "Total Points  : " << fixed << setprecision(2) << semPoints << "\n";
    cout << "Semester GPA  : " << gpa << "\n";
}

double cgpa = overallPoints / overallCredits;

cout << "\n====================================\n";
cout << "           FINAL RESULT\n";
cout << "====================================\n";
for (int i = 0; i < (int)semGPA.size(); i++)
    cout << "Semester " << i + 1 << " GPA : " << fixed << setprecision(2) << semGPA[i] << "\n";
cout << "Total Credits  : " << overallCredits << "\n";
cout << "Overall CGPA   : " << fixed << setprecision(2) << cgpa << "\n";
cout << "====================================\n";
return 0;
}
```

Sample Output

```
====================================
         CGPA CALCULATOR
====================================
Grades: A, A-, B+, B, B-, C+, C, C-, D, F (4.0 scale)

Enter number of semesters: 1
----- Semester 1 -----
Number of courses:
Course 1 name: Grade: Credit hours:
Course 2 name: Grade: Credit hours:
Semester 1 Report
Course                Grade   Credits   Points
-------------------------------------------------
Maths                 A       4         16.00
Physics               B+      3.00      9.90
-------------------------------------------------
Total Credits : 7.00
Total Points  : 25.90
Semester GPA  : 3.70

====================================
           FINAL RESULT
====================================
Semester 1 GPA : 3.70
Total Credits  : 7.00
Overall CGPA   : 3.70
====================================
```